Ipc is the Hosoya index: a numerically stable, size-free reformulation of molecular information content

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Abstract

The information content index `Ipc`, available in standard cheminformatics toolkits, is shown to be both definitionally redundant and numerically unsound. Over 400 molecules, 99.79% of the variance of $\log_2$ Ipc is explained by $\log_2 S$ alone, where S is the Hosoya index; the entropy factor contributes 2.24 bits against 32.3. Separately, the Le Verrier–Faddeev–Frame recursion used to obtain the characteristic polynomial loses all significant digits above approximately 80 atoms: for a 120-atom chain the exact final coefficient is +1 and the computed value is -1.97×10^{26}, without overflow or warning. We give an eigenvalue-based algorithm with magnitude-ordered root expansion, accurate to 2.3×10^{-15} at $n = 200$ and eight times faster. We then define dIpc $= \log_2 Z(G) - \log_2 F(n{+}1)$, matching information relative to the n-alkane of equal size, whose correlation with atom count is +0.049 against +0.993 for $\log_2 Z$. The upper reference $Z_{\max}$ over chemical graphs is determined exhaustively to $n = 16$, by local search to $n = 24$, and by a fitted law accurate to 0.022 bits thereafter. Applied to 6,000 structurally defined terpenes from COCONUT, spanning C10 to C40, against skeleton-size-matched natural products from other pathways, terpene carbon skeletons occupy a quarter of the topological variance (ratio 4.07, Cohen's $d = -1.16$), consistently across skeleton sizes.

Introduction

Molecular descriptor blocks are computed on every molecule in every model that uses them, so a defective descriptor propagates silently and at scale. RDKit's `descList` contains 217 entries, two of which — `Ipc` and `AvgIpc` — are information-theoretic indices over the characteristic polynomial of the adjacency matrix [1,2].

This work asks whether those two indices measure what they are intended to measure, and whether they are computed correctly. Both answers are negative. We identify the quantity `Ipc` actually tracks, give a corrected algorithm for the underlying polynomial, propose a reformulation with analytic references at both ends of its range, and apply it to natural product chemical space.

Methods

Exact characteristic polynomial. The adjacency matrix of a molecular graph is integer, so its characteristic polynomial has integer coefficients. We compute them exactly by running the Faddeev–LeVerrier recursion modulo several small primes and reconstructing by the Chinese remainder theorem, with primes chosen so that every modular matrix product remains exactly representable in double precision (S2.1). Results were verified against exact rational arithmetic.

Information quantities. With coefficient magnitudes c, let $S = \sum |c_i|$ and $H = -\sum (c_i/S) \log_2 (c_i/S)$. The standard descriptors are Ipc $= S \cdot H$ and AvgIpc $= H$. We evaluate both by log-sum-exp so that S is never formed explicitly.

The dIpc index. For an acyclic graph, S equals the Hosoya index Z, the number of matchings [1]. Among trees on n vertices the path maximises Z, with $Z = F(n{+}1)$, the Fibonacci number. We therefore define

$$\mathrm{dIpc}(G) \;=\; \log_2 Z(G) \;-\; \log_2 F(n{+}1),$$

which is zero for an unbranched chain, negative for branched acyclic graphs, and positive when rings contribute additional matchings.

Upper reference. Since adding an edge cannot reduce the matching count, the maximum over chemical graphs (connected, maximum degree ≤ 4) is attained on 4-regular graphs. These were enumerated exhaustively with `geng` from *nauty* [5] and their matchings counted by subset dynamic programming. Beyond $n = 16$ we used local search over 2-opt edge swaps with an exact objective, which yields lower bounds (S2.2); beyond $n = 24$ the dynamic program exceeds memory and we use a fitted law bracketed by an explicit construction below and a density ceiling above.

Natural product data. COCONUT 2022.01.01 structures with NPClassifier pathway labels [6,7,8]. The largest fragment is used throughout, as Z is multiplicative over components.

Structural definition of a terpene. Terpene identity is a property of the carbon backbone, not of the decorated molecule: enzymes add oxidation, epoxidation, acetylation and glycosylation after the skeleton is assembled. We therefore extract the carbon framework — deleting all non-carbon atoms and retaining the largest connected carbon fragment — and require it to satisfy the isoprene rule: a skeleton of 10, 15, 20, 25, 30 or 40 carbons — mono- through tetraterpene (carotenoid) — carrying at least $C/5 - 1$ pendant methyl carbons. This is a structural criterion, independent of the classifier, and dIpc is computed on the skeleton rather than on the parent molecule.

Controls are single-pathway non-terpenoids, matched one-for-one on skeleton carbon count. They are *not* filtered by the isoprene rule: excluding structurally isoprenoid controls leaves an artificially homogeneous residue at small carbon counts (s.d. 0.07, almost entirely linear chains) and reverses the comparison in those bands.

Results

Ipc is dominated by the Hosoya index

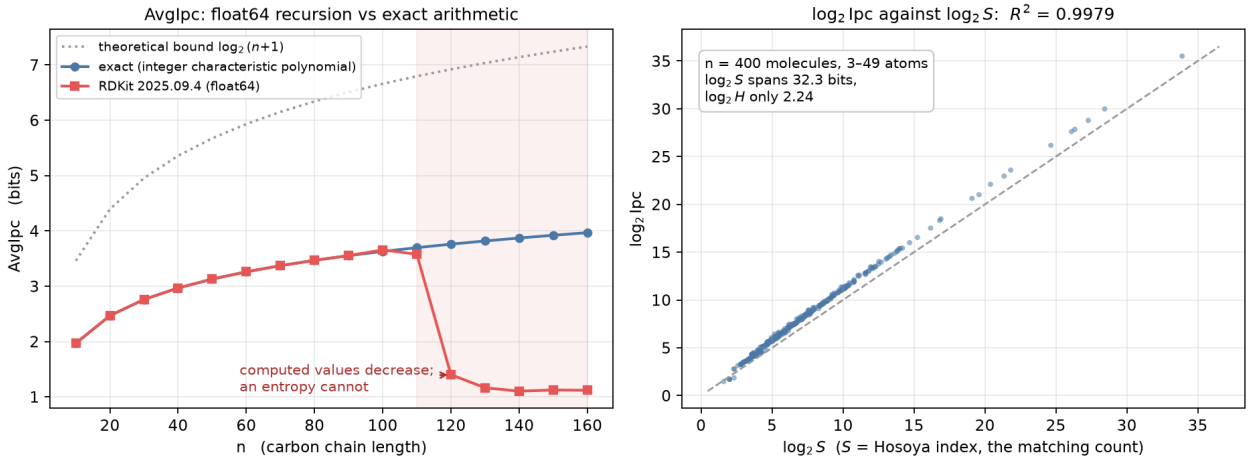


Figure 1: **Left:** `AvgIpc` from RDKit 2025.09.4 compared with exact integer arithmetic for linear carbon chains, $n = 10$–160; dotted line is the bound $\log_2(n{+}1)$. **Right:** $\log_2$ Ipc against $\log_2 S$ for 400 molecules of 3–49 heavy atoms; dashed line is $y = x$.

The Bonchev–Trinajstić total information content $I = N \cdot H$ is defined for a population of N elements partitioned into classes by H [2]. In Ipc $= S \cdot H$, however, H is an entropy over the $n{+}1$ polynomial coefficients, bounded by $\log_2(n{+}1)$, while S counts matchings and grows as φ^n. The two factors refer to different populations.

Empirically the product is dominated by S. Across 400 molecules, $\log_2 S$ spans 32.3 bits and $\log_2 H$ spans

2.24, with $R^2 = 0.9979$ between $\log_2$ Ipc and $\log_2 S$ (Figure 1, right). As $\log_2 S$ correlates with heavy-atom count at $r = 0.993$, `Ipc` is approximately a measure of molecular size.

The characteristic polynomial is computed incorrectly above 80 atoms

The Le Verrier–Faddeev–Frame recursion is sequential: coefficient k is formed after k matrix products have grown the intermediates to the magnitude of the largest coefficient, then obtained as a difference of quantities of that magnitude, with each coefficient entering the next matrix.

n	`AvgIpc`, RDKit	exact	coefficients in error by $> 1\%$
60	3.2594	3.2594	0.0%
80	3.46687	3.46686	—
100	3.6561	3.6278	19.6%
120	1.4007	3.7593	23.0%
250	1.1084	4.2886	—

`AvgIpc` is a Shannon entropy bounded by $\log_2(n{+}1)$ and cannot decrease as a chain is extended; it begins decreasing near 110 atoms (Figure 1, left). Boundedness does not confer safety: the quantity is computed from the same corrupted coefficients and degrades earlier and less visibly than the unbounded `Ipc`. No overflow occurs — at $n = 250$, $S = 1.3 \times 10^{52}$ against a double-precision ceiling of 1.8×10^{308} — so the returned values are finite.

Expanding the polynomial from its eigenvalues, with root factors multiplied in order of increasing magnitude, keeps each partial product near the scale of the final coefficients. Maximum relative error falls from 5.4×10^{15} to 2.3×10^{-15} at $n = 200$, with an eight-fold speed-up. The ordering is essential: unordered expansion still reaches 4.8×10^{4} relative error at $n = 160$. The correction has been submitted upstream (rdkit/rdkit#9657).

dIpc removes the size dependence that other normalisations do not

quantity	correlation with heavy-atom count
$\log_2 Z$	+0.993
$\log_2(Z)/n$	+0.409
dIpc	+0.049

Subtracting $\log_2 F(n{+}1)$ removes the size term exactly, where division by a power of n removes it only approximately. The resulting index is signed and interpretable: zero for an n-alkane, negative with branching, positive with ring fusion. Across 478 acyclic molecules no value exceeded zero, consistent with the extremal property of the path.

Z_max is exact to n = 16 and bounded to 0.022 bits beyond

n	$Z_{\max}$	$\log_2 Z/n$	basis
12	2,921	0.95935	exhaustive
14	11,032	0.95924	exhaustive
16	42,439	0.96082	exhaustive
20	598,516	0.95957	local search
24	8,659,688	0.96025	local search
30	4.69×10^{8}	—	fitted
40	3.66×10^{11}	—	fitted
50	2.86×10^{14}	—	fitted

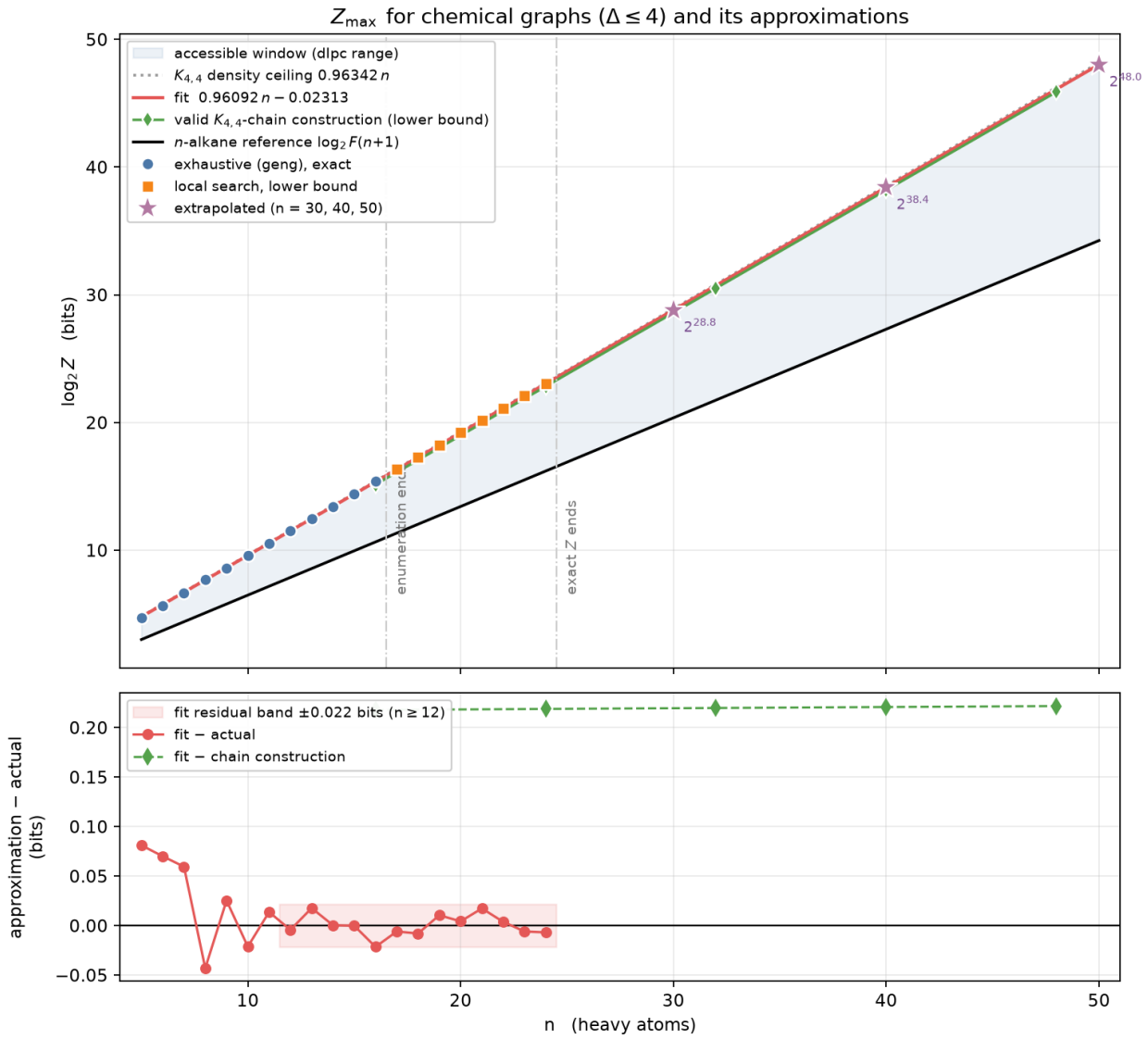


Figure 2: $\log_2 Z$ against n. Blue: exhaustive enumeration, exact. Orange: local search, lower bounds. Green: an explicit $K_{4,4}$-chain construction. Shaded: the interval between the n-alkane reference and $Z_{\max}$. Lower panel: residuals of the two approximations.

The fitted law $\log_2 Z_{\max}(n) = 0.96092\,n - 0.02313$ holds to ± 0.022 bits over $n = 12$–24. It is bounded below by an explicit construction — a chain of $K_{4,4}$ blocks, each with one internal edge removed so that connector vertices remain within degree 4 — whose exact Z is obtained by transfer matrix at any size and which lies a constant 0.22 bits below the fit up to $n = 248$. It is bounded above by the $K_{4,4}$ density ceiling $0.96342\,n$. At $n = 50$ the two bounds differ by 0.25 bits.

The density is maximal at $K_{4,4}$ and the sequence shows period-8 structure: a connected 4-regular graph cannot attain the density of disjoint $K_{4,4}$ blocks, as joining two blocks costs 0.0416 bits.

Terpene skeletons occupy a quarter of the topological variance of other pathways

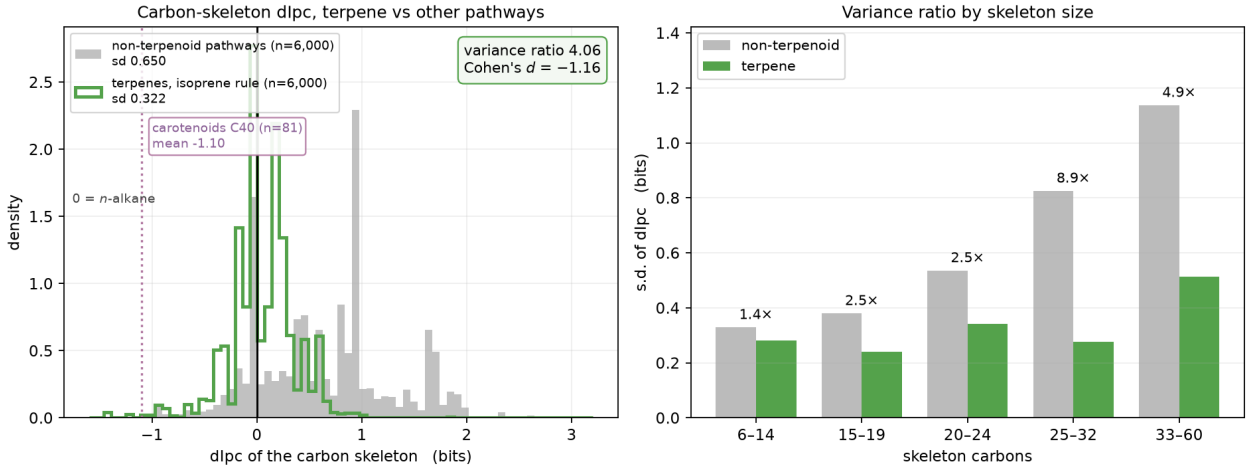


Figure 3: **Left:** dIpc of the carbon skeleton for 6,000 structurally defined terpenes and 6,000 non-terpenoid natural products matched on skeleton carbon count. **Right:** standard deviation by skeleton size, variance ratio annotated.

The isoprene rule is satisfied by 77.2% of NPClassifier-labelled terpenoids and by 11.2% of other pathways, a seven-fold enrichment that provides independent structural support for the labels (S3.3). Carbon counts agree exactly with the predicted class for 71.3% of 11,871 labelled terpenoids and to within three carbons for 87.7%, the remainder being degraded or rearranged skeletons; carotenoids agree best (87.3% exact). Decoration is substantial: the median labelled terpenoid carries seven non-carbon heavy atoms, and monoterpenoids carry more decoration than backbone.

	terpene	non-terpenoid	test
n	6,000	6,000	—
mean skeleton carbons	20.73	20.73	matched exactly
mean dIpc	+0.054	+0.648	Cohen's $d = -1.16$
s.d.	0.322	0.650	Levene $p < 10^{-300}$
variance ratio	—	4.065	—

Terpene skeletons occupy approximately a quarter of the topological variance of size-matched natural products from other pathways, and lie close to the n-alkane reference while other pathways lie well to the ring-fused side. Variance ratios by skeleton size are 1.37, 2.50, 2.46, 8.90 and 4.88 over 6–14, 15–19, 20–24, 25–32 and 33–60 carbons, covering the full terpene range from monoterpene to carotenoid.

Class means separate within the terpenes. Carotenoids are the extreme: a C40 skeleton carrying eight methyl branches gives mean dIpc -1.10 with s.d. 0.218 ($n = 81$), the most branched and the most topologically uniform of any class (S3.3).

The effect strengthens as the measured object approaches the biosynthetic one:

definition	variance ratio	Cohen's d
whole molecule, classifier labels	2.045	−0.86
carbon skeleton, classifier labels	3.223	−0.96
carbon skeleton, isoprene rule	4.065	−1.16

Removing enzymatic decoration and restricting to skeletons that satisfy the isoprene rule each increase the separation, which is the behaviour expected if the constraint acts on the backbone rather than on the decorated product.

Discussion

`Ipc` should be removed from descriptor blocks. It is approximately a function of molecular size, which is already represented by numerous other descriptors, and it is computed from a polynomial that is unreliable above 80 atoms. Models trained on molecules below about 60 atoms are unaffected by the correction, since old and corrected values agree to 10^{-15} in that range; above it, feature distributions change and should be re-examined rather than simply recomputed.

`AvgIpc` should be retained once the underlying polynomial is corrected, and `dIpc` added alongside it. The cost of dIpc is approximately 10 ms per molecule at $n = 50$ using the exact path. It contributes no additional information to a model already provided with $\log_2 Z$ and n and sufficient flexibility to combine them (partial $r = +0.0095$); its value lies in interpretability, in extrapolation beyond the training size range, and in settings where the model cannot learn the interaction. For a feature matrix, $\log_2 Z$ and n are better supplied as separate columns than pre-combined: division by $n^{1/3}$, $n^{2/3}$ or n adds nothing once both are present (partial r of −0.087, −0.092, −0.096).

Terpene biosynthesis leaves a measurable signature in graph topology. Assembly from a single branched C5 unit confines terpene carbon skeletons to a quarter of the topological variance of skeleton-size-matched natural products from other pathways, holding them near the unbranched chain reference while other pathways extend to the ring-fused side. The effect is large ($d = -1.16$) and uniform in direction across skeleton sizes from C10 to C40.

That it strengthens monotonically as decoration is removed and as the terpene definition is tightened from a predicted label to a structural criterion indicates that the constraint acts on the backbone, as the biosynthesis implies, rather than on the elaborated product. dIpc is consequently usable as a size-independent descriptor of biosynthetic class, not only as a repair of `Ipc`.

Limitations

The isoprene rule applied here is a necessary condition, not a decision procedure: a carbon count divisible by five with the expected methyl density admits some non-terpenoid skeletons and excludes rearranged or degraded terpenes, including steroids and apocarotenoids. The 23% of labelled terpenoids it rejects are not a random sample. Carotenoids and sesterterpenes are represented by 81 and 72 skeletons respectively, so their class means are the least precise.

Control composition is consequential. Filtering controls to exclude structurally isoprenoid molecules inverts the comparison below 20 carbons, because what survives at those sizes is almost entirely linear chains (s.d. 0.07); controls are therefore defined by pathway alone. COCONUT 2022.01.01 is a literature aggregation and no deduplication was applied beyond fragment selection, so related scaffolds may be over-represented in both groups.

$Z_{\max}$ values for $n = 17$–24 are lower bounds rather than proven maxima, with a validated worst-case shortfall of 0.48%; the fitted law is correspondingly a slight underestimate. The optimality of $K_{4,4}$ is confirmed exhaustively at $n = 8$ and $n = 16$ but its asymptotic form is taken from the literature.

dIpc contributes no additional information to a model already supplied with $\log_2 Z$ and n and able to combine them nonlinearly; its value is interpretability and extrapolation, not predictive gain. Its components are established prior results [1,4]; a literature search over OpenAlex returned no previous definition of $\log Z - \log F(n{+}1)$ as a descriptor, but this is not an exhaustive novelty search.

References

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Supplementary information

S1. Data

source	count	range	use
boiling-point set	1,000 molecules	2–63 heavy atoms	general chemical space
— variance decomposition subset	400	3–49	Figure 1
COCONUT 2022.01.01, NPClassifier-labelled	407,029	—	terpenoid pool
— labelled terpenoids, skeleton validation	11,871	—	S3.3
— terpenes passing the isoprene rule	6,000	10–40 skeleton C	Figure 3
— pathway controls, skeleton-size-matched	6,000	10–40 skeleton C	Figure 3
connected 4-regular graphs	8,037,418 at n=16	$n = 5$–16	$Z_{\max}$

Terpenoid superclass counts in COCONUT: diterpenoids 23,082; triterpenoids 22,280; sesquiterpenoids 20,418; steroids 16,768; monoterpenoids 6,207; meroterpenoids 3,607; sesterterpenoids 937; carotenoids 953.

S2. Algorithms

S2.1 Modular characteristic polynomial

Primes are selected such that $n(p-1)^2 < 2^{53}$, which keeps each modular matrix product exactly representable in double precision and allows the inner loop to use a floating-point matrix multiply rather than arbitrary-precision integer arithmetic. The number of primes is set from a Hadamard bound on the coefficient magnitudes. Output was verified identical to Faddeev–LeVerrier over exact rationals on random sparse symmetric graphs up to $n = 30$.

S2.2 Local search over 4-regular graphs

Initialisation is a circulant graph $C_n(1, 2)$, which is 4-regular and connected. Each step proposes a 2-opt edge swap: edges (a, b) and (c, d) are deleted and (a, c), (b, d) inserted, leaving all vertex degrees unchanged. Swaps that disconnect the graph are rejected; the exact matching count is recomputed and the swap accepted when it does not decrease. Multiple restarts from independent random states are used. Every reported value is the exact Z of a realised graph and is therefore a lower bound on $Z_{\max}$.

Validation against exhaustively known maxima: recovered at $n = 10$, 12, 13 and 15; at $n = 14$ the best of 24 restarts fell 0.48% short.

S2.3 Block-chain transfer matrix

For blocks joined in a path by single edges, with per-block counts N, N_l, N_r, N_{lr} (total matchings, and those leaving the left, right, or both connectors unmatched),

$$u_{i+1} = u_i N_{lr} + (u_i + m_i)N_r, \qquad m_{i+1} = u_i(N_l - N_{lr}) + (u_i + m_i)(N - N_r),$$

where u and m count configurations with the right connector unmatched and matched. This is exact and linear in the number of blocks. An initial version used intact $K_{4,4}$ blocks, which raises connector degree to 5 and violates the valence constraint; one internal edge is therefore removed per block.

S3. Supplementary results

S3.1 Timings

step	n	wall time
exhaustive enumeration and exact Z	10	0.03 s
	12	0.07 s
	14	4.3 s
	15	52 s
	16	1,140 s
local search, 2,000 moves	20	2.1 s
	24	42 s
block-chain transfer matrix	≤ 248	0.17 s

Exhaustive enumeration scales at roughly 10–20× per added atom; $n = 17$ would require approximately six hours and $n = 18$ approximately three days.

S3.2 Exhaustive $Z_{\max}$

n	5	6	7	8	9	10	11	12	13	14	15	16
$Z_{\max}$	26	51	100	209	388	780	1482	2921	5600	11032	21482	42439

Generated graph counts match OEIS A006820. The maximum at $n = 8$ is $K_{4,4}$. At $n = 10$ the maximum over all connected graphs of maximum degree ≤ 4 (89,402 graphs) equals the maximum over 4-regular graphs, consistent with edge monotonicity.

S3.3 Structural validation and class breakdown

Carbon-skeleton size against the size implied by the predicted class, 11,871 labelled terpenoids:

class	expected C	n	exact	within 1	within 3	median decor.
Monoterpenoid	10	2,482	58.0%	78.1%	85.5%	11
Sesquiterpenoid	15	2,500	83.5%	89.7%	92.3%	5
Diterpenoid	20	2,500	74.3%	83.5%	87.2%	7
Sesterterpenoid	25	936	60.8%	76.6%	87.1%	5
Triterpenoid	30	2,500	67.0%	78.8%	82.6%	14
Carotenoid	40	953	87.3%	90.8%	96.5%	4
all		11,871	71.3%	82.7%	87.7%	7

Decoration is the count of heavy atoms outside the carbon framework; 2.1% of labelled terpenoids are undecorated and the 90th percentile is 30.

SMARTS substructure frequencies, labelled terpenoids against other pathways ($n \approx 6{,}000$ each):

pattern	SMARTS	terpenoid	other	enrichment
isopropenyl	`[CH3][CX3](=[CH2])[#6]`	5.6%	0.7%	7.9×
trisubstituted alkene	`[CH3][CX3](=[CX3])[#6]`	37.2%	7.0%	5.3×
gem-dimethyl	`[CH3][CX4]([CH3])[#6]`	56.7%	11.5%	4.9×
isoprene C5 frame	`[CH3][#6]([#6])[#6][#6]`	92.7%	28.6%	3.2×
isopropyl	`[CH3][CHX4]([CH3])[#6]`	12.7%	6.2%	2.1×

The composite rule — skeleton carbons in $\{10, 15, 20, 25, 30, 40\}$ with at least $C/5 - 1$ pendant methyls — is satisfied by 77.2% of labelled terpenoids and 11.2% of other pathways.

dIpc of the carbon skeleton, by class, within the 6,000 terpenes passing the rule:

class	n	mean skeleton C	mean dIpc	s.d.
Monoterpene C10	441	10.7	−0.099	0.302
Sesquiterpene C15	1,886	15.4	+0.110	0.261
Diterpene C20	1,963	20.1	+0.098	0.339
Sesterterpene C25	72	25.1	−0.329	0.451
Triterpene C30	1,557	29.7	+0.054	0.217
Carotenoid C40	81	39.9	−1.101	0.218
all terpenes	6,000	20.7	+0.054	0.322

S3.4 Normalisation exponents

Partial correlation with the boiling-point residual for $\log_2(Z)/n^k$, as the regression basis is enriched:

basis	$k = 1/3$	$k = 2/3$	$k = 1$
size terms only, linear	+0.444	+0.426	+0.360
size $+ \log_2 Z$	+0.247	+0.220	+0.060
size $+ \log_2 Z +$ nonlinear n	−0.087	−0.092	−0.096

S4. Software and reproducibility

Code, data and figures: **github.com/guillaume-osmo/kumomlx**

component	path
exact characteristic polynomial, dIpc, $Z_{\max}$	`src/kumomlx/ipc.py`
derivation and measurements	`DESCRIPTORS.md`
upstream patch and Eigen C++ implementation	`rdkit-fix/`
$Z_{\max}$ enumeration	`rdkit-fix/zmax/`
COCONUT analysis	`paper/coconut.py`, `paper/coconut_stats.py`
figures	`paper/fig_*.py`

RDKit 2025.09.4; nauty 2.9.3, invoked as `geng -c -d4 -D4 n`.